# INSY 7130 TEST 2 COMPLETE NOTES – PAGE 1 OF 2

**05a EMBEDDINGS • 06b–08a CLUSTERING • REPRESENTATION, FITTING, DIAGNOSTICS, VALIDATION**

## COURSE LOGIC + REPRESENTATION

- Driving question: can structure in observations help a decision, and can it survive credible challenges? Start with uncertainty, useful pattern, method/choices, validation, and action. Evidence supports a claim; an algorithmic result is not proof.
- Neighbors answer "which past cases resemble this one?" Clustering organizes the whole set. Anomaly detection prioritizes departures. Time series studies ordered change/forecasting.
- Numbers define similarity. Nominal integer codes invent order/gaps; one-hot makes all category mismatches equally distant. Ordinal codes preserve order but may imply equal spacing. Feature choice, scaling, weights, and metric determine geometry.
- Euclidean $d(x,y) = \sqrt{\sum_j (x_j - y_j)^2}$; cosine $= x^T y/(\|x\|\|y\|)$. For unit vectors, dot product = cosine. High semantic similarity can support retrieval but not necessarily an operational claim (e.g., negation).

## 05a: DIMENSION + MANIFOLDS

- Ambient dimension = supplied measurements; intrinsic = local degrees of freedom; output = requested coordinates. Asking for a 2-D map does not establish intrinsic dimension 2.
- A manifold is locally describable by ordinary coordinates even if globally bent. Linear projection can collapse separate folds; neighbor paths may trace curvature, but noise, sparse sampling, and nearby folds can create false connections.
- Rotation/reflection/translation changes coordinates but preserves Euclidean distances/neighbors. Compare relationships, not orientation.

## PCA, t-SNE, UMAP

- PCA: shared linear directions maximizing retained variance; scores are projections; loadings and explained-variance ranking exist; fitted PCA can transform new rows. It can lose nonlinear neighborhood structure.
- t-SNE: convert input distances to directional Gaussian neighbor probabilities $p_{j|i}$; adjust each width to a requested perplexity; symmetrize pairs; initialize output (often PCA); use heavy-tailed Student-$t$ output weights $w_{ij} = 1/(1 + \|y_i - y_j\|^2)$; iteratively minimize mismatch using attraction when $p > q$ and repulsion when $q > p$.
- Perplexity = equivalent number of equally likely neighbors; larger means broader input neighborhoods, not more/fewer clusters. Exact layout depends on seed, initialization, optimization, and settings. sklearn `TSNE.fit_transform`; no `transform` for future rows.
- UMAP: find input neighbors → weighted local graph → optimize low-dimensional layout. `n_neighbors` changes graph breadth; `min_dist` changes output packing, not graph edges and is not a strict pairwise minimum. `umap-learn` can transform future rows. **Map traps:** Clear islands, sizes, densities, and global separations do not prove natural clusters or faithful global distances. t-SNE/UMAP axes have no direct feature meaning.

## EMBEDDING VALIDATION

- Top-$k$ retention for row $i$: $|N_k^{in}(i) \cap N_k^{map}(i)|/k$; average across rows and inspect individual records. "Best" depends on what must be preserved and whether future placement is required.
- Check original images/text, reasonable settings, seeds, samples, and independent data. Stability is useful but does not prove relevance.
- Label-agreement statistic: mean fraction of each row's fixed input neighbors sharing its label. Permutation null: labels exchangeable while vectors/neighbors stay fixed. One-sided Monte Carlo $p = (1 + \#\{T_b \geq T_{obs}\})/(B+1)$. Preserve writers/entities/blocks if exchangeability requires restricted shuffling.
- Small $p$ supports label association in original representation, not ten natural clusters, a correct map, or a best method.

## K-MEANS OBJECTIVE + FIT

- Request $K$; initialize centers; assign each row to nearest center; replace each center with assigned mean; repeat until center movement $<$ `tol` or `max_iter`. Each assignment/update reduces or preserves $W = \sum_i \|x_i - \mu_{c(i)}\|^2$ (inertia/SSE).
- `k-means++` favors initial rows far from chosen centers; `random` samples rows. Later centers are means and need not be observed. `n_init` runs complete fits and retains lowest final inertia, not best initial inertia or most frequent partition. `random_state` reproduces random choices; defaults in course version: $K = 8$, `max_iter=300`, `tol=.0001`, algorithm Lloyd.
- Learned: `labels_`, `cluster_centers_`, `inertia_`, `n_iter_`. `predict` assigns nearest fitted centroid; `transform` returns distances to centers. Reuse fitted preprocessing.
- `StandardScaler`: $z = (x - \bar{x})/s$. Scaling changes nearest centers/boundaries; it prevents numerical-unit dominance but does not prove equal substantive importance.
- Example $\{1, 2, 8, 9\}$, start centers 1,2: inertia 85; settled centers 1.5,8.5, inertia 1. **Limits:** Forces every row into a group; favors compact/roughly spherical groups; outliers pull means; labels/colors arbitrary; $K = 3$ may split one wide population rather than reveal a third population; near-equal inertia can encode very different operational stories.

## CHOOSING K

- Inertia cannot select $K$: it never increases as centers are added. Elbow seeks diminishing improvement and can be ambiguous.
- Silhouette: $a(i)$ mean distance to own cluster; $b(i)$ smallest mean distance to another; $s(i) = [b - a]/\max(a, b)$. Near 1 separated; near 0 overlap; negative closer to another cluster; undefined for $K = 1$. Mean silhouette rewards compact separated groups and may undervalue curves.
- Gap: $\text{Gap}(k) = B^{-1}\sum_b \log W_{kb}^* - \log W_k$; $s_k = SD(\log W_{kb}^*)\sqrt{1 + 1/B}$. Choose first $k$ with $\text{Gap}(k) \geq \text{Gap}(k+1) - s_{k+1}$. Depends on no-group reference; bars are simulation allowances, not CIs; not a $p$-value or proof of true $K$.
- If scores are close, prefer simpler $K$ unless an added group is stable, interpretable, and changes a useful action. Inspect members/profiles/original features/new period.

## HIERARCHICAL CLUSTERING

- Agglomerative: one cluster/row; repeatedly merge closest current pair under linkage; record children, height, size; continue to one. Greedy/irreversible; early merges cannot be undone; a cut need not globally optimize its cluster count.
- Single=min cross-pair distance (connected/elongated; chaining); complete=max (keeps all members relatively close); average=mean cross-pair; Ward=smallest increase in within-cluster SSE (Euclidean; compact).
- Example $d(A,C) = 3.16$, $d(B,C) = 2.24$: single 2.24, average 2.70, complete 3.16. Merge height is linkage value, not every within-cluster pair distance.
- Dendrogram leaves=observations, vertical joins=linkage values; horizontal leaf spacing is layout only. Follow branches. Horizontal cut gives flat groups. Cuts of one tree are nested; separate K-means fits need not be.

## HIERARCHICAL SOFTWARE/CUTS

- SciPy `linkage(X,method,metric)` returns $(n-1) \times 4$ rows: child 1, child 2, height, number of original rows. Original IDs $0 \ldots n-1$; new IDs start at $n$. `dendrogram` draws existing history.
- `fcluster(...,criterion="distance")` uses $t$ as height. `maxclust` finds a threshold producing at most $t$ clusters; ties may give fewer. `cut_tree(...,n_clusters=k)` follows merge order to request a count. Other criteria: `inconsistent`, `monocrit`, `maxclust_monocrit`.
- Fixed data/order, representation, metric, linkage, and implementation are deterministic, though tied distances can admit different merge orders. Cluster labels are arbitrary.
- No built-in `predict`/decision boundary for future rows in ordinary agglomerative clustering; add a separate rule or refit/reassess.

## DBSCAN

- Density-Based Spatial Clustering of Applications with Noise. `eps`=radius; points exactly at radius count. `min_samples`=required neighborhood count including self.
- Core: count $\geq$ minimum. Border: non-core within `eps` of a core. Noise: neither, sklearn label $-1$. Clusters grow through connected cores, then absorb adjacent borders. **Border is not a bridge:** it joins but cannot extend to its neighbors.
- Larger `eps`: more neighbors, less noise, possible merging. Smaller: fragmentation/more noise. Larger `min_samples`: harder core status/more noise. Scaling/metric change neighborhoods.
- Strength: irregular/curved connected regions and explicit noise; endpoints can be farther than `eps` via core chains. Weakness: one global density rule struggles with groups of unequal density. Seek a stable parameter range, not one attractive image.
- Learned: `labels_`, `core_sample_indices_`. Noise may be error, rare legitimate case, or parameter artifact. No built-in sklearn prediction rule for new rows.

## GAUSSIAN MIXTURE MODELS

- $p(x) = \sum_{g=1}^{G} \pi_g \mathcal{N}(x|\mu_g, \Sigma_g)$, $\pi_g \geq 0$, $\sum_g \pi_g = 1$. Posterior $P(g|x) = \pi_g \mathcal{N}_g(x)/\sum_h \pi_h \mathcal{N}_h(x)$.
- EM: initialize means/covariances/weights; E-step membership probabilities; M-step weighted parameter updates; repeat until likelihood improvement is small/limit. Multiple starts retain highest likelihood. Check `converged_`, `n_iter_`.
- `predict_proba` preserves soft membership; `predict` gives argmax hard label. A 0.516/0.484 label is ambiguous. High relative membership can occur where all components have low absolute density.
- Covariance: full=own tilt/shape/size; tied=one shared covariance; diag=own axis-aligned variances; spherical=one variance/component. For $p$ features, full has $p(p+1)/2$ values/component; diag $p$; spherical 1; tied one shared full matrix.
- Extra components may approximate skew/non-Gaussian shape rather than identify populations. Ellipses show component spread/equal density, not hard boundaries or confidence intervals. Fitted weights need not equal hard-label fractions.

## METHOD SELECTION

- K-means: compact center-based groups + future nearest-center assignment. Hierarchical: nested resolutions/merge history. DBSCAN: irregular dense connectivity + noise. GMM: overlapping elliptical distributions + uncertainty. Method output never establishes causality, a "real" population, or useful action by itself.

## CLUSTER VALIDATION + INFERENCE

- Internal geometry (inertia/silhouette/gap) is not truth/usefulness. Inspect members, profiles, geometry, added distinctions, settings/features/starts/samples, external outcomes, another period, and domain knowledge.
- ARI compares pairwise co-membership ignoring label numbers: 1 identical; about 0 chance agreement; negative below chance. Subsampling (e.g., 80% rows) checks sensitivity within observed data, not independent replication. Initialization and observation stability answer different questions.
- rapDouble dipping: clustering on $x$, then ordinary $t$-testing cluster means on $x$, reuses the same information; K-means constructed separation. Tiny naive $p$ can arise from one Gaussian population. Simple holdout assignment on the same features may still create differences by construction; shuffling final labels omits cluster selection.
- Valid reference comparison: predefine features/statistic/null; simulate complete datasets under justified reference; repeat all preprocessing, clustering, count/feature search in each; compare observed statistic. Example ratio $W_{K=2}/W_{K=1}$, smaller = stronger split. Parametric bootstrap from fitted one-Gaussian reference includes refitting K-means.
- Monte Carlo tail $(1 + \#\{T_b \text{ as/extremer}\})/(B+1)$. Evidence against one Gaussian does not prove exactly two populations; skew/heavy tails may also reject reference. Formal test needs calibrated false-positive behavior.
- If many methods/settings were tried and only the strongest reported, simulate the search or use fresh evaluation data. Several geometry scores are not independent confirmation.

## HIGH-YIELD CLUSTER TRAPS

- Cluster IDs/colors have no stable cross-fit meaning; compare memberships/profiles. Stable/separated $\neq$ causal/real/useful. Changing representation, scale, distance, linkage, $K$, density settings, covariance, or reference changes the fitted evidence. A later threshold only selects from fixed scores.
- K-means and GMM use starts; DBSCAN outputs count; hierarchy builds once then cuts; K-means retains lowest inertia; GMM highest likelihood. K-means/ordinary hierarchy assign all observed rows; DBSCAN may label noise; GMM probabilities cover all components.

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09b–10a ANOMALY DETECTION • 11a–12a TIME SERIES, DECOMPOSITION, DEPENDENCE, FORECASTING

## ANOMALY FOUNDATIONS

- A detector converts a chosen meaning of unusualness into a score. Reference defines unusual: mixed collection vs new readings against accepted healthy baseline (novelty). Check load/design/environment comparability.
- Point anomaly: one row departs. Contextual: unusual only within condition/regime. Collective: relationship/joint pattern is unusual even when individual values are familiar.
- SCORE = method-specific unusualness; RULE = threshold/rank/budget; FLAG = selected for follow-up; FAULT/OUTCOME = independently established condition. Do not equate them.
- Workflow: define target/reference → features/scale → fit → orient scores → choose rule from costs/capacity → investigate records/context/sensors → evaluate external outcomes/new data.

## LOF FORMULAS + INTERPRETATION

- For point $p$, $N_k(p)$ excludes self. $r_k(o)$ is the distance from neighbor $o$ to its $k$th-nearest other point.

$$\text{reach}_k(p,o) = \max\{d(p,o), r_k(o)\}$$

$$\text{lrd}_k(p) = \left[\frac{1}{|N_k(p)|}\sum_{o\in N_k(p)} \text{reach}_k(p,o)\right]^{-1}$$

$$\text{LOF}_k(p) = \frac{|N_k(p)|^{-1}\sum_{o\in N_k(p)} \text{lrd}_k(o)}{\text{lrd}_k(p)}$$

- Reachability is directional because receiving neighbor $o$ supplies its radius. It raises short distances to neighbor scale. LOF near 1=comparable local support; > 1=lower local density than neighbor average. Absolute sparsity alone need not be locally unusual.
- Seven values $0, 1, 3, 10, 50, 60, 80$, $k = 2$: for 10, neighbors 3,1; reaches 7,9; lrd=.125; neighbor lrds .400,.333; LOF=2.933. For 80, reaches 20,30; lrd=.04; neighbor comparison gives LOF=.917. Changing to $k = 3$ moves priority toward wide group (about 2.718) and 10 falls to .953.
- $k$ defines comparison scale and can reverse ranking. Scaling, features, and metric change neighborhoods. sklearn stores negative LOF: larger-unusual convention is `-negative_outlier_factor_`. `kneighbors()` returns ordinary distances; reachability occurs internally.
- For unseen rows fit `LocalOutlierFactor(novelty=True)` on accepted reference; transform with reference scaler; score unseen only. Novelty calls do not reproduce training-collection scores.

## ISOLATION FOREST

- In each current node choose a feature randomly, then a threshold uniformly between that feature's current min/max; split and repeat. Root/path/leaf; path length counts splits from zero. Wide adjacent gaps/endpoints provide many isolating cuts.
- Forest averages randomized trees, often built from row samples. Short average path = easy isolation = high anomaly score after sample-size normalization; score near .5 roughly matches reference path length, not 50% fault probability. Leaves may contain multiple rows; scoring estimates remaining path.
- `n_estimators`=trees; `max_samples`=rows/tree; `random_state`=repeatability. More trees reduce dependence on one random sequence. A row can be scored through a tree even if not sampled to build it.
- sklearn `score_samples`: lower = more anomalous; use negative for larger-unusual. LOF asks "less locally supported than neighbors?" Isolation Forest asks "easy to isolate?" Thus 10 leads LOF while endpoint 80 leads isolation.
- LOF distances usually need deliberate scaling. Isolation Forest splits one feature at a time; linear unit changes preserve corresponding split positions, so standardization is generally unnecessary.

## THRESHOLDS, BUDGETS, METRICS

- Lower threshold with larger-unusual scores can only preserve/add flags; workload and recall cannot decrease; precision may rise or fall. Scores/ranking stay fixed. Do not share a numerical cutoff across LOF and isolation scales.
- Fixed budget: rank and review top $N$; cutoff varies by collection. Fixed numerical threshold: workload varies. State tie rule because score ties may select $> N$. `contamination` sets a training cutoff/expected fraction; in lecture implementations it changes flags/`offset_`, not raw training scores or reference membership.
- TP=flag+outcome; FP=flag+no outcome; FN=no flag+outcome; TN=no flag+no outcome. Precision $= TP/(TP + FP)$; recall $= TP/(TP + FN)$; specificity $= TN/(TN + FP)$; FPR $= FP/(FP + TN)$; $F_1 = 2PR/(P + R)$.
- Checking only flagged cases can estimate precision, but recall needs total positives or representative review of unflagged cases. Three flags do not imply three faults; unusual healthy cases may reveal a new regime.

## 10a WINE EXAMPLE + VALIDATION

- UCI data: 1,599 red and 4,898 white wines; 11 chemistry features + sensory quality. Use red reference; 150 mg/L total sulfur dioxide exceeds about 99.4% of red but only 63.1% of white values. Pooling obscures context. Quality excluded from detector inputs.
- Standardize chemistry; LOF $k = 20$; Isolation Forest 300 trees, `max_samples=256`. At budget 50, respective cutoffs about 1.506 and .554; both flag 50, only 19 overlap. Detector choice changes investigations.
- Define low rating $\le 4$: 63/1599. LOF top 50 gives TP=7, FP=43, FN=56, TN=1493; precision 14%, recall 11.1%. Increasing LOF review to 100 adds 3 low-rated +47 others: precision 10%, recall 15.9%. Low rating is an external proxy, not identical to chemical abnormality/fault.
- Random-review null: choose 50 distinct records uniformly; expected low-rated $50(63)/1599 = 1.97$. Hypergeometric $P(X \ge x) = \sum_{j=x}^{\min(n,K)} \binom{K}{j}\binom{N-K}{n-j}/\binom{N}{n}$. For $x = 7$, exact one-sided $p \approx .00275$ (20,000 simulation estimate .00265). Supports association for fixed procedure, not causation, every flag, completeness, optimality, or future generalization.
- Sensitivity: LOF $k = 10$ vs 20 at budget 50 changes 17 selections; low-rated found falls 7 to 4. Compare methods at equal workload; freeze chosen procedure before fresh evaluation.

## CONTAMINATION + MASKING

- Reference contamination: target-abnormal behavior influences fitted reference. Masking: similar abnormalities support each other and become less unusual.
- Demo: keep background + focal remote row fixed; remote group size 5 → 30 with LOF $k = 20$: focal LOF about 6.19 → 1.01 because all 20 neighbors can become remote peers. LOF rank 5/205 →125/230; Isolation Forest rank 2/205 →25/230.
- More flags/changed `contamination` do not cure masking because raw relationships are unchanged. Use accepted-normal reference, regime-specific comparisons, provenance checks, robust features, temporal holdout, and sensitivity to reference composition; investigate groups of anomalies.

## NEW-DATA EVALUATION

- Freeze features, scaler, reference model, and budget/threshold; score separate collection without outcomes; select; then reveal outcomes and compare with same-size random review. Keep repeated/related records together. If only reviewed outcomes exist, estimate precision but sample unreviewed rows to assess misses/recall.
- Isolation Forest scores new rows through fixed trees. LOF needs `novelty=True`. Scoring preserves reference; refitting updates it and can introduce masking. Decide deliberately when new behavior enters reference.

## TIME INDEX + MOVING AVERAGES

- Parse with explicit date format; sort; require unique labels, no missing values, and expected complete calendar (e.g., `date_range(...,freq="QS"/"MS")`). A row lag equals a time lag only with verified spacing. Dates may label intervals (quarter start), not point measurements.
- `.loc["1992"]` selects a year; partial date slices include endpoints. `shift(h)` aligns earlier values; initial unavailable pairs are NaN.
- Trailing $m$-MA uses current/prior rows. Centered MA uses observations before/after and produces endpoint NaNs. Example centered 3-MA $(443 + 410 + 420)/3 = 424.33$. Centered windows use future values and leak in real-time forecasting.

## ADDITIVE DECOMPOSITION

- $y_t = T_t + S_t + R_t$ (trend-cycle, seasonal, remainder, original units). Fitted $\hat{y}_t = T_t + S_t$; $R_t = y_t - \hat{y}_t$. Reconstruction is exact by remainder definition, not evidence of forecast accuracy/causality.
- Manual classical workflow: moving-average trend; detrend; average departures by season; repeat seasonal effects; remainder. Trailing annual MA lags and loses first $m-1$ rows. Classical `seasonal_decompose(y,model="additive",period=12)` uses centered trend, fixed seasonal effects adjusted to mean zero, and endpoint gaps (monthly even period: six at each end).
- Fixed seasonality, complete-window loss, and outlier influence are limitations. Components estimated on full history are descriptive historical fits, not forecasts.
- STL = Seasonal and Trend decomposition using Loess; alternates local smoothers. `STL(y,period=12,seasonal=13,robust=True)`: period=cycle length; odd seasonal window smooths across same-month occurrences; robust reweighting reduces influence of large remainders. Endpoints estimated with less-balanced local support. Flexible components are not uniquely true.
- Seasonal adjustment: $y_t^{adj} = y_t - S_t = T_t + R_t$; keeps original units/level and unusual departures while removing recurring pattern. It is not seasonal differencing.

## LAGS + ACF

- Lag $h$ compares $y_t$ with $y_{t-h}$; for complete quarterly data, lag 4=same quarter prior year. Lag plot x=earlier, y=current. Seasonal positive ACF often appears at multiples of period; opposite seasons may be negative; trend can create persistent positive ACF.
- Sample ACF $r_h = \dfrac{\sum_{t=h+1}^{T}(y_t-\bar{y})(y_{t-h}-\bar{y})}{\sum_{t=1}^{T}(y_t-\bar{y})^2}$. One full-series mean/denominator differs slightly from Pearson correlation on shortened pairs. ACF measures linear dependence.
- White noise: mean 0, constant finite variance, zero ACF at all nonzero lags; iid is sufficient, not necessary. Approx pointwise 95% bounds $\pm 1.96/\sqrt{T}$; many inspected lags increase chance crossings. Inside bands/zero ACF does not prove independence.

## STATIONARITY + DIFFERENCING

- Stationary: joint distribution unchanged by time shift; changing mean, spread, or fixed seasonality violates it. Stationary series may still be dependent. Constant expected level is insufficient: random walk $y_t = y_{t-1} + e_t$ has variance growing with time; levels share accumulated history.
- Ordinary difference $\Delta y_t = y_t - y_{t-1}$ = change per observation interval; `diff()` loses first row. Random-walk differencing recovers increments; current level is simple forecast if expected next increment is zero.
- Seasonal difference $\Delta_m y_t = y_t - y_{t-m}$; monthly annual `diff(12)` loses first 12. Seasonal adjustment subtracts estimated $S_t$ and retains level; seasonal difference subtracts observed prior season and produces change.
- Log seasonal difference $\log y_t - \log y_{t-m} = \log(y_t/y_{t-m})$; .10 means ratio $e^{.10} = 1.105$, about 10.5% growth. Logs can stabilize proportional seasonal swings.
- Inspect time plot + ACF after transformation. Remaining ACF can be useful stationary dependence. Unnecessary differencing can introduce dependence. Second ordinary difference is change in first differences, not lag-2 difference.
- Ljung–Box null at $H$: population ACFs at lags $1 \ldots H$ jointly zero. `acorr_ljungbox(x,lags=[10])` gives one joint test through 10. Small $p$ rejects; large $p$ means compatible, not proof of independence/white noise (Google changes example $p \approx .637$).

## FORECAST DESIGN

- Chronological split: training ends before test. In lecture, train through June 2019; test July 2019–June 2024 (60 months). Every data-dependent step must fit on training only.
- Baseline: time-only `LinearRegression` on elapsed month, ignoring seasonality. A more complex forecast must improve on the same withheld dates.
- Classical component forecast: decompose training only; fit linear regression to available trend-cycle; extrapolate month numbers; repeat training seasonal averages; add trend + seasonal. Assumes trend and fixed seasonal pattern persist; training trend $R^2 \approx .35$; five-year MAPE about 3.81% in example, with sustained post-2019 level miss.
- `STLForecast(training,ARIMA,order=(1,1,0),trend="t",period=12, seasonal=13, robust=True)`: STL seasonality + supplied model for adjusted series + seasonal forecast. Not equivalent to classical+linear trend; both decomposition and extrapolation differ.
- Error $e_t = y_t - \hat{y}_t$.
- MAE $= n^{-1}\sum |e_t|$ original units;
- RMSE $= \sqrt{n^{-1}\sum e_t^2}$ emphasizes large misses;
- MAPE $= 100n^{-1}\sum |e_t/y_t|$, undefined at zero/unstable near zero.
- Compare methods on identical withheld observations. Average error hides bias, timing, seasonal misses, and degradation by horizon. Evaluate windows (months 1–12, 13–24, 25–60), multiple origins, and relevant horizons. One split does not establish universal superiority.

## LEAKAGE + FINAL TRAPS

- Leakage: full-data imputation/scaling/PCA/ decomposition/feature selection; centered rolling features; later-value gap filling; post-decision inspection/repair fields; target proxies hidden in inputs; overlapping sensor windows/repeated entities across split; repeated tuning against test. Fit the complete pipeline inside available training history; group related records; after test-driven changes obtain a fresh final evaluation. **Final reminders:** Identify reference/units/time interval; distinguish evidence-changing settings from flag thresholds; read score direction; preserve preprocessing; inspect original records; report what evidence does not prove; do not confuse fit/reconstruction with prediction or association with causation.